The Media Connectivity Association



OVERVIEW AND UPDATES · IMMERSIVE MEDIA

Beyond 2D Video Experiences

Evaluating video formats beyond 2D: stereoscopic, multi-view plus depth, point clouds and dynamic meshes

3GPP

OCTOBER 2026 – www.5g-mag.com

LEARN

About the project

The documents the project rests on

3GPP TR 26.956

Technical Report

Evaluation and Characterization of Beyond 2D Video Formats and Codecs, the output of the study FS_Beyond2D.

3GPP TS 26.501

and TS 26.512

The 5G Media Streaming architecture and protocols the formats are assessed against.

ISO/IEC 23090-5

Standard

Visual Volumetric Video-based Coding (V3C), referenced for evaluation.

FIND OUT MORE **Technical Analysis** **Specifications**

LEARN

Formats under evaluation

The four format families of the study, with the coding approaches evaluated for each

Stereoscopic video a left and right view: a fixed 3D effect, no free viewpoint

MV-HEVC | Simulcast or frame-compatible HEVC (anchor)

Multi-view plus depth several views with depth: a limited change of viewpoint

MIV profile of V3C | MV-HEVC

Dense dynamic point clouds a volumetric capture of an object or person

V-PCC | G-PCC, noted as a related technology

Dynamic meshes a volumetric representation with time-varying textured 3D meshes

V-DMC

The report also surveys formats under research (neural radiance fields, light fields, 3D Gaussian splatting) without evaluating them.

LEARN

Services evaluated

The three services of the study, as listed by the reference repository

Stereoscopic

Live, device to device

UE-to-UE stereoscopic video live streaming. Planned in the repository.

Point cloud

On demand

Streaming of professionally produced volumetric video, a single asset containing people.

Multi-view

Plus depth, on demand

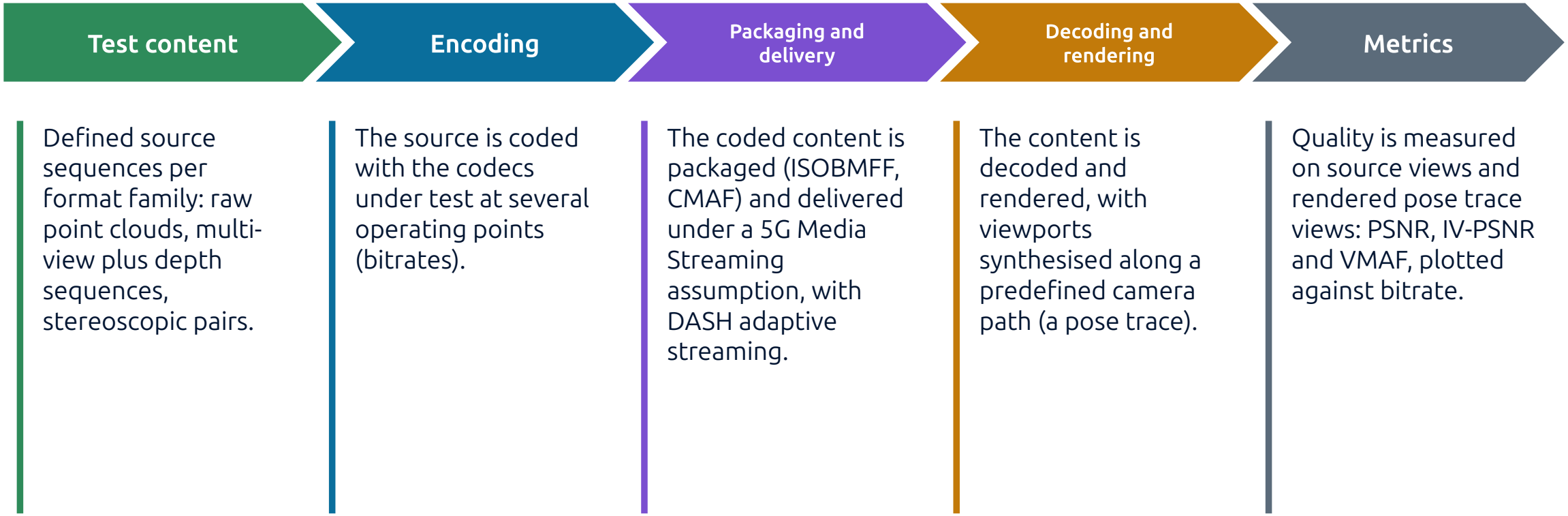
On-demand streaming of multi-view plus depth.

FIND OUT MORE **Technical Analysis** **Specifications**

LEARN

Evaluation pipeline

A repeatable pipeline, so that results are comparable across formats and contributors

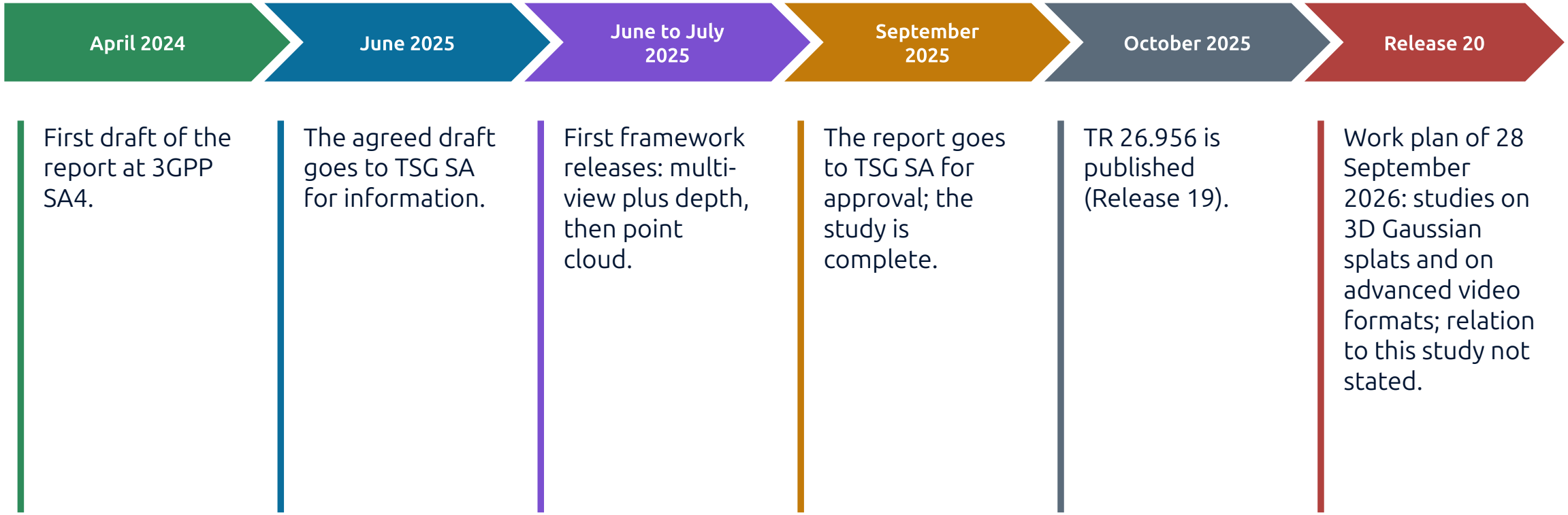


Sources, bitstreams and rendered pose trace videos are large, so the framework references content on external servers.

LEARN

How it grew

The 3GPP study and the 5G-MAG evaluation framework, from the first draft to Release 20



Related Release 19 work: VOPS adds TS 26.265, with stereoscopic operation points based on MV-HEVC.

IMPLEMENT

Repository

The one repository of the evaluation framework, from its README

REPOSITORY	WHAT IT IS	COVERS	LATEST RELEASE	
rt-beyond2d-evaluation-framework	scripts and configuration files for the reproducible evaluation of FS_Beyond2D: multi-view plus depth (TMIV) and V-PCC point clouds	Multi-view plus depth, Point cloud	v0.2.1 24 Jul 2025	GitHub

FIND OUT MORE Technical Analysis Specifications

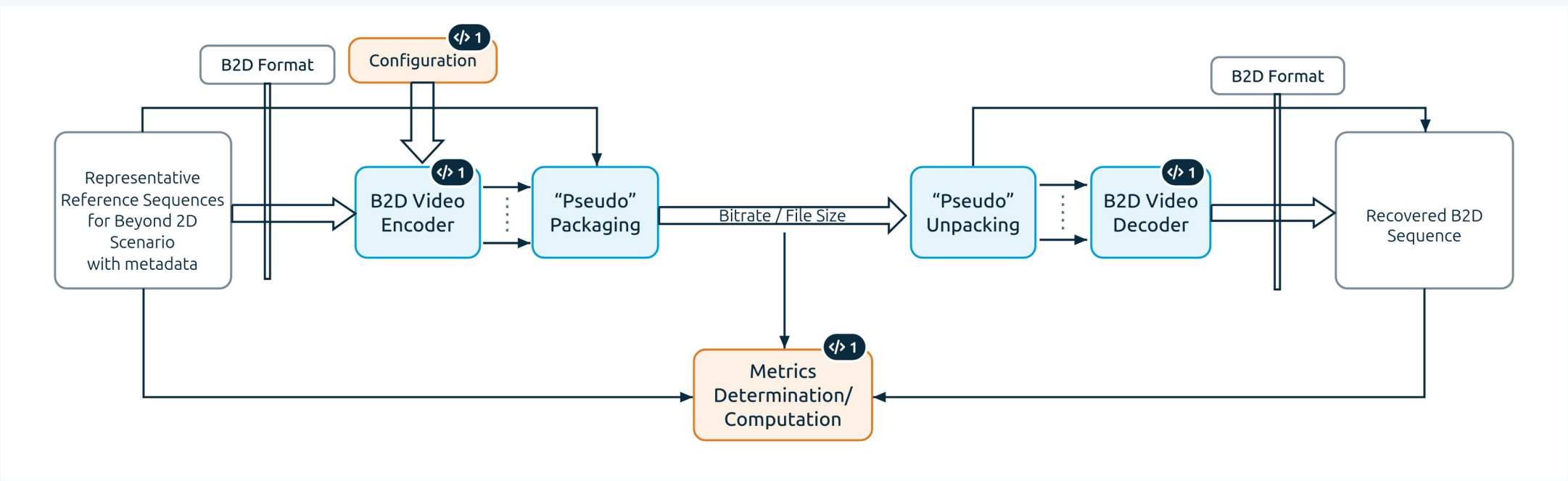
● released ○ no release yet

More information at 5g-mag.com/tech/beyond-2d

IMPLEMENT

Repositories in the architecture

The Reference Tools on the specification figure, as the figure generator draws them



PROVIDED BY THE REFERENCE TOOLS

1 **rt-beyond2d-evaluation-framework**

Mapped from the repository's MVD scenario only; boxes without a badge have no repository.

CONTRIBUTE

Latest updates and roadmap

Status on 22 October 2026

RELEASED

rt-beyond2d-evaluation-framework v0.2.1

24 Jul 2025. Adds the point cloud setup and updates for video generation scripts.

IN DEVELOPMENT

Evaluation services

Completing the evaluation setups for the three services of the study.

PLANNED

Test content

Collecting the initial contents for the evaluation of the FS_Beyond2D formats.

MERGED SINCE V0.2.0 1 pull request in 1 repository

The Media Connectivity Association

See it. Try it. Contribute.

Raise an issue, send a pull request or join the monthly public call.

GitHub

5g-mag.com

Become a contributor →